Week 14

Paper Before Product

SoW, Pricing & Solution Architecture

Your mindset this week: Consultant — scope it, price it, and de-risk it before anyone builds.

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Your roadmap for today — where you are, and what's next

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TODAY'S CLASS SoW, Pricing & Architecture — four phases, start to finish

① Estimate
▸ you are here

- Milestones & sizing
- T-shirt → ranges
- Risk buffers

② Price

- Pricing models
- Sierra & Palantir
- Price the pilot

③ Architect

- Reference architecture
- Box by box
- Stack-choice matrix

④ Survive procurement

- The interrogation
- Name the controls
- Milestone M1

★ **M1 — SoW + Architecture + Pricing**

Where we are: Scope → SELL → Build → Integrate → Prove

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Stage	The question	Week
Scope	What is the real problem, measurably?	W13 ✓
Sell	What will it cost, and what are we buying?	**W14 — today**
Build	Does it work on real cases?	W15
Integrate	Can it see the client's systems, safely?	W16
Prove	Can we prove it, then hand it over?	W17

The through-line: M0 success criteria → M1 acceptance criteria → W17 eval suite. Nothing you wrote in discovery is throwaway.

Today's Agenda

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1. **The Reveal** — M0 consequences + the CFO memo
2. **Estimation & SoW** — milestones, T-shirt sizing, risk buffers; SoW essentials
3. **Pricing models** — T&M-cap / fixed / outcome — Sierra & Palantir
4. **Reference architecture** — the Meridian diagram, box by box; stack-choice matrix
5. **Procurement interrogation** — the centerpiece — answer the security questionnaire live
6. **Integration topology** — P2P / iPaaS / event mesh
7. **War story & M1** — close on the CISO's synthetic-data memo

Terms you'll hear today

Term	In plain words
Procurement	the buyer's vetting + contracting gate — legal, security, and finance deciding whether to sign.
SoW (Statement of Work)	the contract that scopes the build: deliverables, acceptance, price, IP, change control.
Acceptance criteria	the measurable test for 'done' — your M0 success criteria, written into the contract.
T&M / fixed / outcome	the three ways to bill: time-and-materials (per hour), fixed (one price), or per business result.
OBO (on-behalf-of)	the agent acts under a specific user's identity, not a standing super-user login.
RBAC	role-based access control — who can see/do what, enforced by role.
MCP server	one governed 'door' every tool call passes through — where access is enforced and audited.
RAG	retrieval-augmented generation — ground answers in your policy docs, with citations.
iPaaS / event mesh	integration patterns: a managed hub (iPaaS) or a pub/sub backbone (event mesh).

First principles: why paper before product?

- **The contract decides whether there is a build** — not the demo. Scope, price, and controls are the gate.
- **The deal is won by people who can say no** — and most of them — procurement (the buyer's vetting gate) and the CISO — are never in the demo.
- **Acceptance criteria = the success criteria from discovery** — so 'done' is objective, not a taste fight.
- **The paper you write this week is the build plan** — every SoW promise becomes a W15–17 deliverable.

The Reveal — what M0 bought you

Three anonymized student charters on screen. We critique them against the through-line.

CFO memo: "Board meets in 6 weeks. Send me cost, timeline, and what exactly we are buying — by Friday."

The trap: quote a number to look decisive. The FDE move: scope first, then commit — but visibly, so you don't look like you're stalling.

Chat-storm #1

TYPE **Quote a number, or buy time? Which, and why?**

The nuance: buy time on the number, but commit to a DATE for the number, and commit to milestones now. A board-facing CFO can't hear 'it depends.'

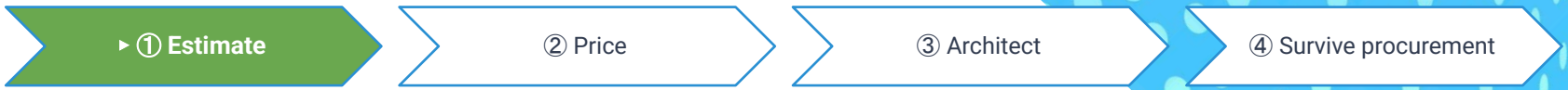
Phase 1 of 4

Estimation

From a vague ask to a defensible number.

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▸ Now starting **phase 1 of 4 — Estimate**



▸ **This phase's activity:** Activity: size three milestones

Estimation: from vague ask to defensible number

- **Milestones** decompose the engagement into shippable units with sign-off gates (M0→M4 already do this).
- **T-shirt sizing** S/M/L/XL per milestone instead of false-precision hours; convert to ranges.
- **Risk buffers** multiply by an uncertainty factor PER risk class — not one flat 20%.

The output is a RANGE with a rationale — never a single number you'll be held to. Name what would move you to the top or bottom.

▶ ACTIVITY

Size three milestones

2 minutes — give each a T-shirt size and one reason. No hours allowed.

You try it first — then we reveal how an FDE would do it.

Activity: size three Meridian milestones

DO 2 minutes, in chat. Give each a T-shirt size (S / M / L / XL) and one reason. No hours allowed.

- **The triage agent** classify a dispute as Type-A (fraud) vs Type-B (merchant).
- **The MCP integration layer with RBAC** one governed door across 4 systems, scoped per analyst.
- **The full eval suite + CI gate** 150-case golden set gating every deploy.

Resist false precision. 'It's a Large because it spans four systems and a security review' beats any hour count. Paste your three sizes; we reveal the key next.

Reveal: T-shirt sizing → defensible ranges

Size	Range	Use for (Meridian examples)
S	3–5 days	a single tool wrapper; a policy-doc loader
M	8–12 days	the triage agent; the evidence gatherer
L	15–25 days	the MCP integration layer with RBAC
XL	30–50 days	the full eval suite + CI gate + dashboards

Then apply a risk multiplier per class. The board-ready number is low / expected / high, defensible line by line. (Calculator: notebook §1.)

8–12 weeks

of data-access politics for every **1 week** of value demo. (Palantir reality.)

- **It's not a build risk — it's an organizational-access risk.** The longest pole in a regulated client is permission to touch the data, not the code.
- **For Meridian, the CISO controls review is the gate.** Budget for the politics, or eat them. 'Data-access approval delay' is the top risk-register line.

Source: Nabeel Qureshi, 'Reflections on Palantir' — 'data is political.' The CISO IS that pole.

Engineer economics: project the cost before you quote

Two numbers every FDE carries: what an engineer earns, and what a solution costs to build.

- **Revenue per engineer** ~1 FDE covers 3–5 enterprise accounts at ~$200K each → roughly $0.6–1.0M of revenue per engineer per year. That's the number the business runs on.
- **Project the build cost upfront** your sizing (days) × a blended day-rate = the engineering cost of THIS solution. An FDE puts a number on it BEFORE building — not after the overrun.
- **Defend the gap** the price has to clear cost + the access-politics buffer + margin. If the pilot can't, you scope it smaller or walk — you don't discover it mid-build.

An FDE who can't cost their own solution can't price it. Estimation isn't busywork — it's how you protect the margin that keeps the role funded.

SoW essentials — the four things procurement reads first

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SoW = Statement of Work — *the contract that scopes the build. Procurement reads these four sections first.*

Section	What it nails down	Failure if you skip it
Acceptance criteria	measurable done, tied to M0 eval anchors	endless 'is it done?' fights
IP terms	who owns agents, prompts, fine-tunes, embeddings	lawsuit at renewal
Data annex	what data, where, retention, deletion, subprocessors	CISO blocks the deal
Change control	how scope changes get priced	scope creep eats your margin

Acceptance criteria = M0 success criteria. That's why you scoped evals at discovery — acceptance is objective, not a taste fight.

Quick Poll #2

Q Which SoW section will procurement attack hardest at Meridian?

(a) Acceptance criteria

(b) IP terms

(c) Data annex

(d) Change control

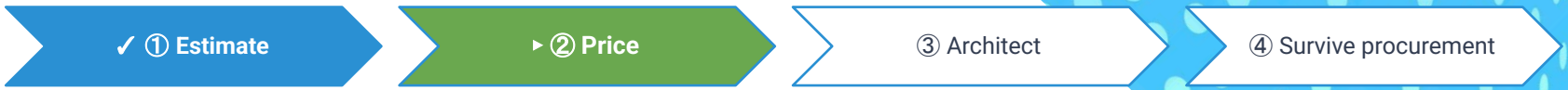
Phase 2 of 4

Pricing

Pick a model, defend it, price the pilot.

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✓ Finished phase 1 (Estimate) · ▸ Now phase 2 of 4 — Price



▸ **This phase's activity:** Decision poll: price the pilot

Pricing models — pick one, defend it

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Model	You bill on	Risk sits with	Margin
T&M with a cap	hours, ceiling-bounded	shared (client caps exposure)	predictable, capped upside
Fixed-price	a scoped deliverable	VENDOR (you eat overruns)	high if scoped well, brutal if not
Outcome-based	a business result	shared, results-linked	highest upside & variance

The question procurement asks every model: 'Who counts the outcomes / the hours / done?' Whoever owns that definition owns the invoice.

Case study · Sierra — outcome-based pricing

- **What** Sierra (Bret Taylor & Clay Bavor) prices customer-service AI agents on the RESOLUTION — not seats or hours.
- **Scale** reported $150M+ ARR with outcome-based pricing as the headline model.
- **Why it works** the agent's value IS the resolved ticket, so billing the resolution aligns vendor and client.
- **Why it's hard** you need an airtight, agreed definition of 'resolved' and instrumentation both sides trust.

Meridian parallel: 'resolved dispute' is exactly such an outcome — but is it resolved when the agent drafts, when the customer accepts, or when no chargeback follows?

Case study · Palantir — acquire, expand, scale

- **The motion** land small, embed, solve one urgent workflow, prove value, then expand across the account.
- **The arc** the Airbus engagement became Skywise — an engagement that turned into a product.
- **The reality underneath** 8–12 weeks of access politics per week of value demo — so the pilot is priced to survive the access fight; expansion is where economics work.

Synthesis: Sierra shows you CAN price on outcomes when the outcome is clean; Palantir shows the early reality is access-dominated — so price the pilot conservatively and bet on expansion.

Price the Meridian pilot — three ways

Same scope (triage + evidence + policy + comms agents, 60k disputes/yr, 10-week pilot), three commercial wrappers:

- **T&M-cap** blended rate × estimated days, capped — predictable, protects margin, limited upside.
- **Fixed-price** one number for the milestone set — you win if discovery was good.
- **Outcome-based** base + bonus per dispute auto-resolved within Reg E — aligned, variable.

Recommendation for a first pilot with a nervous regulated client: FIXED-PRICE FLOOR + OUTCOME BONUS.

The estimation & pricing calculator

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Bring your own numbers: size the milestones, project the cost, and compare the four pricing options live.

▶ CALCULATOR

`https://ik-fde-calculator.vercel.app`

(live — open it now in a second tab)

- **Estimate** size each milestone → days → cost (low / expected / high).
- **Price** four models — incl. the recommended fixed-floor + outcome bonus — with a margin check.

Use it live during the price-the-pilot poll — defend a number, not a vibe.

Quick Poll #4 — the decision poll

Q Price the Meridian pilot:

(a) T&M with a cap

(b) Fixed-price

(c) Outcome-based

(d) Fixed floor + outcome bonus

Phase 3 of 4

Solution Architecture

The diagram you put in front of the CISO

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✓ Finished phase 2 (Price) · ▸ Now phase 3 of 4 — Architect



▸ **This phase's activity:** Architecture walk + chat-storm

Solutions first: a building block for each problem

Before the diagram: take each problem from discovery and name the technical solution. The architecture is just these, wired together safely.

The problem (from discovery)	The technical solution
Classify Type-A (fraud) vs Type-B (merchant)	a Triage agent
Evidence is scattered across 4 systems	tools behind one governed MCP door
Answers must cite Reg E / GLBA correctly	RAG (retrieval over the policy knowledge base)
Money-moving actions can't be automated blind	a human-in-the-loop approval queue
Act as the right analyst, fully audited	identity (OAuth2 OBO) + RBAC in the access layer
'Prove it' before it ships	an evaluation gate in CI

Pick the solution per problem first; the next slide stacks them into layers, and the slide after is all of it wired together.

Build it in layers — each layer solves a class of problems

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- **Layer 1 · Systems of record** the bank's four systems (core banking, ServiceNow, SharePoint, warehouse) — the data, untouched.
- **Layer 2 · Governed access** the MCP door + identity (OBO) + RBAC — the ONE place every tool call passes through, audited.
- **Layer 3 · Agents** Triage · Evidence · Policy · Comms — the reasoning, which never touches the systems directly.
- **Layer 4 · Grounding & proof** RAG over policy + an eval gate — answers that are correct and provable.
- **Layer 5 · Human-in-the-loop** the approval queue for money-moving actions — the safety net.

Stack these five layers and you get the reference architecture on the next slide — each layer a section you can defend on its own.

The Meridian reference architecture

CLIENT (Meridian) systems

- Core banking (REST)
- ServiceNow (tickets)
- SharePoint (policy)
- AWS IAM / Identity Center

MCP server
one governed door · OAuth2 OBO · audit

AWS — THE CLIENT'S CLOUD (Bedrock, client account)

Agents: Triage · Evidence · Policy · Comms
OpenAI Agents SDK · Claude on Bedrock

- Bedrock Knowledge Base (RAG)
- Eval gate in CI (Promptfoo)
- OpenTelemetry → CloudWatch (cost/latency/eval)
- **Ops analyst UI — human-in-the-loop queue**

Agents never touch systems directly — every tool call funnels through the one governed MCP door. RBAC is enforced in that deterministic layer, never in the LLM.

Box by box — built to pass security

Box	What / stack	Why it's there
Triage/Evidence/Policy/Comms	OpenAI Agents SDK · Claude on Bedrock (client's AWS)	the workflow is genuinely multi-agent
Custom MCP server	one governed door, multi-tenant, OAuth2 OBO	agents never touch systems directly; enforce scope + audit
Bedrock Knowledge Base	policy-doc RAG with citations	grounds answers in Reg E/GLBA policy
AWS IAM + scoped agent keys	identity for users AND agents	agent acts AS the analyst; least privilege
Promptfoo eval gate in CI	continuous eval + CI deploy gate	'prove it' becomes a number; acceptance = thresholds
Analyst HITL queue	human approval surface	money-moving actions need a human

Stack-choice matrix — and when the answer flips

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Dimension	OpenAI SDK + AWS Bedrock (ours)	Azure + Foundry	Google ADK + Gemini
Client already on	AWS (most banks are)	Microsoft / Azure / Entra	Google Cloud / Workspace
Enterprise identity	IAM + Identity Center; AgentCore Identity	Entra Agent ID (native)	Workspace IAM
Compliance posture	strong: VPC isolation, GLBA-mature on AWS	SOC2/GLBA mature on Azure	strong, GCP-native
When it flips	— (Meridian runs on AWS)	client is M365/Entra-first	client is GCP-first; cost-sensitive

The honest line: we picked our stack because Meridian already runs on AWS. Stack choice follows the client, not your preference — that maturity is what interviewers reward.

Chat-storm #2

TYPE **Which box breaks first at scale — and why?**

- **The MCP server** every tool call funnels through it — throughput chokepoint + SPOF (the cost of one governed door).
- **The legacy core banking** 1990s middleware, real rate limits — won't care that you have agents.
- **Eval token cost** running 150-case golden evals on every commit gets expensive (W17's lesson).

Break · 15 minutes

Back in 15. After the break: the procurement interrogation — six of you answer the security questionnaire live, as the vendor.

Have the 15-question questionnaire open. The pass bar: name a concrete control. The fail bar: 'we'll figure that out during the build.'

Phase 4 of 4

The Procurement Interrogation

Answer the security questionnaire — live, as the vendor

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✓ Finished phase 3 (Architect) · ▸ Now phase 4 of 4 — Survive procurement

✓ ① Estimate → ✓ ② Price → ✓ ③ Architect → ▸ ④ Survive procurement

▸ **This phase's activity:** The procurement interrogation

Procurement interrogation — the rules

SCENARIO **Meridian's procurement committee + the CISO are on the call. They have your SoW and a 15-item security questionnaire. You're the vendor. They can kill the deal today.**

1. **Setup** — the pre-recorded procurement + CISO (the CISO) videos play.
2. **Pick** — instructor selects 6 of the 15 questionnaire items.
3. **Answer** — a named student answers AS the vendor, in chat, within 2 minutes.
4. **Pressure** — instructor plays the hostile follow-up live until the soft spot shows.
5. **Debrief** — was that a control, or a slide answer?

You attempt all six rounds first — THEN we reveal what a passing answer sounds like. Pass bar: name a concrete control. Fail bar: 'we'll figure that out during the build.'

▶ ACTIVITY

Answer the questionnaire

Six live rounds — you're the vendor on the procurement call. Name the control.

You try it first — then we reveal how an FDE would do it.

Activity: answer the questionnaire — six live rounds

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One named student per row answers AS the vendor, in chat, in 2 minutes. The instructor plays the hostile follow-up. Attempt it before you see the model answer.

#	Questionnaire item	What you're being tested on
3	Synthetic vs production data?	do you protect the bank's data by default?
4	Whose identity does the agent act under?	least-privilege / on-behalf-of identity
8	How is every action audited?	accountability + traceability
9	What about prompt injection?	adversarial-input defense
10	What if it states the wrong Reg E deadline?	handling confidently-wrong output
15	Who owns the IP, and exit?	commercial + offboarding terms

Don't name the control yet — answer as you would on the call. We grade the gap between your answer and the reveal next.

Reveal: what a passing answer sounds like

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Q (#4): *'Whose identity does your agent act under?'*

Weak: 'The agent has secure access to your systems.'

Strong: 'The agent acts under the analyst's identity via OAuth2 on-behalf-of. It never has standing access. Scope is enforced in the MCP layer, not the model, and every call is audited with the acting identity. An analyst scoped to Type-A disputes physically cannot pull Type-B records through the agent — and we'll prove it with the two-analyst test.'

A passing answer names the control, names where it lives (integration layer, not the LLM), and offers proof.

Reveal: the control each question wanted

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#	Questionnaire item	The control a passing answer names
3	Synthetic vs production data?	synthetic-only until controls review passes
4	Whose identity does the agent act under?	OAuth2 OBO; scope in MCP, audited
8	How is every action audited?	append-only audit log w/ acting identity
9	What about prompt injection?	input guardrail; quarantine poisoned chunks
10	What if it states the wrong Reg E deadline?	citation tripwire; HITL on low confidence
15	Who owns the IP, and exit?	IP annex; data deletion + exit plan

Score the room: for each question, did anyone name THIS control? Every one of these you'll BUILD in W15–17 — that's the payoff of paper-before-product.

Integration topology — P2P / iPaaS / event mesh

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Pattern	What it is	Wins when	Hurts when
Point-to-point	direct system-to-system	few systems, simple, fast	N systems → N² connections
iPaaS	managed integration hub	many systems, governance needed	cost, lock-in, latency
Event mesh	pub/sub event backbone	real-time, decoupled flows	operational complexity; overkill for req/reply

For the pilot: few systems, request/reply, ship in 10 weeks → P2P through the one MCP door. Revisit the event mesh when the account expands to event-driven intake.

"Best engineering loses to a weak data annex."

A real engagement that turned on the security questionnaire or the pricing model — not the technology.

The lesson FDEs relearn constantly: the deal is decided by the people who can say no — and most of them are not in the demo.

Section check — all four phases complete

✓ All 4 phases complete

✓ ① Estimate → ✓ ② Price → ✓ ③ Architect → ✓ ④ Survive procurement

★ You produced the M1 package — SoW + Architecture + Pricing

You priced and architected the whole engagement on paper. One memo left — and it changes how you build.

Milestone M1 & the cliffhanger

- **2-page SoW** acceptance criteria = your M0 eval anchors.
- **Architecture diagram + stack-choice matrix**
- **Pricing model with rationale**
- **Risk register** (access-politics on top)
- **AWS sandbox access confirmed** (the class Bedrock keys; fallback: --mock mode needs nothing).

the CISO's memo: *"Security review: no production data until controls review passes. Build against synthetic data only."*

Sources — verify these

- Sierra outcome-based pricing & $150M+ ARR — Bret Taylor / Clay Bavor; Sierra public statements & press (2024–2025).
- Palantir 'acquire, expand, scale' + 8–12 weeks access-politics per week of value + 'data is political' — Nabeel Qureshi, 'Reflections on Palantir'; Airbus → Skywise public material.
- Pricing triad (T&M-cap / fixed / outcome) — standard professional-services commercial models; risk-transfer is industry-standard.
- Stack facts (OpenAI Agents SDK + AWS Bedrock preferred per PM 2026-06-30; AgentKit/Assistants API deprecations) — program stack statement.

THANK YOU

W14 — Sell. Next: we build.



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